

(Following paper code and roll no. to be filled by student at the answer books.)

PAPER CODE: BMIC 104

ROLL NO.

BBS COLLEGE OF ENGINEERING & TECHNOLOGY
PRAYAGRAJ

SECOND SESSIONAL EXAM (2025-2026)

COURSE: MCA

SEM: 1st

SUBJECT: Discrete Mathematics

MAX MARKS: 70

MAX TIME: 2:00 HR

NOTE: 1. QUESTION PAPER CONTAINS THREE PARTS.
2. ALL SECTIONS ARE COMPULSORY.

(SECTION A)

[7 X 2 = 14]

Q1. ATTEMPT ALL FROM THE FOLLOWING:

- State and prove the recurrence relation for $S = \{1, 1, 2, 3, 5, 8, 13, \dots\}$.
- Show that the statements: $P \rightarrow Q$ and $\neg Q \rightarrow \neg P$ are equivalent.
- If $A = \{1, 2, 4\}$, $B = \{2, 5, 7\}$ and $C = \{1, 3, 7\}$ find $(A \cap B) \cap (A \cap C)$.
- Consider a ring $(R, +, *)$ defined by $a * a = a$. Determine whether the ring is commutative or not.
- Let R be a relation on R , the set of real numbers, such that $R = \{(x, y) : |x - y| < 1\}$ is an equivalence relation.
- Find the solution of $a_n - 4(a_{n-1} - a_{n-2})$, with initial condition $a_0 = a_1 = 1$.
- Simplify: $F(A, B, C, D) = \sum(0, 1, 4, 5, 6, 8, 9, 12, 13, 14)$

(SECTION B)

[5 X 7 = 35]

Q2. ATTEMPT ANY FIVE OF THE FOLLOWING:

- Let G be the set of all non-zero real numbers and let $a * b = ab/2$. Show that $(G, *)$ is an abelian group.

- How many arrangements can be made by using the letters of the word "COMPUTER" so that arrangements begin and ends with vowel?
- Show Prove that $6^{n+2} + 7^{2n+1}$ is divisible by 43 for each positive integer n .
- State and prove Lagrange's theorem.
- For the set $I_4 = \{0, 1, 2, 3, 4\}$. Show that the modulo 4 is a field.
- How many different rooms are needed to assign 500 classes, if there are 45 different time periods during in the university time table that are available?

(SECTION C)

[3X7= 21]

Q3. ATTEMPT ANY THREE OF THE FOLLOWING:

- What do mean by valid arguments? Proof the validity of argument- For students to do well in discrete structure course: it is necessary that they study hard. Students who do well in courses do not skip classes. Students who study hard do well in courses. Therefore students who do well in discrete structure course do not skip class.
- Express the following statements using predicate and quantifiers:
 - For every student in this class, that student has studied mathematics.
 - Some student in this class has visited United States.
 - Every student in this class visited either Canada or United States.
- Let $G = \{1, -1, i, -i\}$ with the binary operation multiplication be an algebraic structure, where $i = \sqrt{-1}$ be an abelian group. Also find generator of G .
- Solve $a_{n+2} + 2a_{n+1} - 15a_n = 6n + 10$

(Following paper code and roll no. to be filled by student at the answer books)

PAPER CODE: BMC-106	ROLL NO.								
---------------------	----------	--	--	--	--	--	--	--	--

BBS COLLEGE OF ENGINEERING & TECHNOLOGY

PRAYAGRAJ
SECOND SESSIONAL EXAM (2025-2026)

COURSE: MCA

SEM: 1st

SUBJECT: Cyber Security

MAX MARKS: 70

MAX TIME: 2:00 HR

NOTE: 1. QUESTION PAPER CONTAINS THREE PARTS.
2. ALL SECTIONS ARE COMPULSORY.

(SECTION A)

[7 X 2 = 14]

Q1. ATTEMPT ALL FROM THE FOLLOWING:

- Explain the term Security Risk Analysis.
- Discuss the difference between malware and spyware.
- Explain the working of CCTV.
- List out the various need for an Information Security Policy.
- What do you mean by Risk Management?
- Write short notes on ISO.
- Illustrate the term Software Piracy.

(SECTION B)

[5 X 7 = 35]

Q2. ATTEMPT ANY FIVE OF THE FOLLOWING:

- Illustrate the various types of threats in an Information Security? Explain the concept of CIA. Fraud also.
- Discuss about Intrusion Detection Systems. Explain its types in detail.
- Explain the followings malwares: - Virus, Worms, Trojan-horses and Backdoors.
- For RSA algorithm, calculate Ciphertext (C) for the given plaintext input of M = 88 and the two prime numbers, p = 17 and q = 11.
- Why security policies should be developed? Explain the Policy Review Process also.
- Discuss about Cyber Law. State some Cyber Laws in India.

(SECTION C)

[3X7= 21]

Q3. ATTEMPT ANY THREE OF THE FOLLOWING:

- What are the data security considerations? Explain in the reference of Data Backup security, data archival security and data disposal considerations.
- Discuss the Security issues in Hardware, Data Storage & Downloadable Devices.
- Discuss the publication and notification requirements of policies. Also explain the various types of security policies.
- What do you mean by Intellectual Properties rights (IPR). Write short notes on Patents, Copyrights and Trademarks.

(Following paper code and roll no. to be filled by student at the answer books)

PAPER CODE: BMC103
ROLL NO.

BBS COLLEGE OF ENGINEERING & TECHNOLOGY

PRAYAGRAJ
SECOND SESSIONAL EXAM (2025-2026)

COURSE: MCA SEM: 1ST
SUBJECT: PRINCIPLES OF MANAGEMENT & COMMUNICATION
MAX MARKS: 70 MAX TIME: 2:00 HR

NOTE: 1. QUESTION PAPER CONTAINS THREE PARTS.
2. ALL SECTIONS ARE COMPULSORY.

(SECTION A)

[7 X 2 = 14]

Q1. ATTEMPT ALL FROM THE FOLLOWING:

- Explain management.
- Name the key steps involved in the planning process.
- List the key principles of directing.
- Define Motivation.
- Differentiate between centralisation and decentralisation.
- Discuss the concept of pitch.
- State the elements of communication process.

(SECTION B)

[5X7=35]

Q2. ATTEMPT ANY FIVE OF THE FOLLOWING:

- Discuss the various types of leadership style.
- Discuss Maslow's and Herzberg's two factor theory of Motivation.
- 'Use of technology makes our communication effective.' Illustrate.
- Discuss the type of flow of communication in an organisation.
- Discuss in detail the types, structure and relevance of report.
- What is controlling in management? Discuss its nature, importance and limitations.

(SECTION C)

[3X7= 21]

Q3. ATTEMPT ANY THREE OF THE FOLLOWING:

- Explain communication barrier, its types and ways to overcome it.
- Discuss the principles and techniques of Scientific Management.
- Evaluate the format of business letter with all the entries in detail.
- Summarize the Hawthorne Experiments conducted by Mayo.

[illegible]

PRAYAGRAJ
SECOND SESSIONAL EXAM (2025-2026)

NOTE: 1. QUESTION PAPER CONTAINS THREE PARTS.
2. ALL SECTIONS ARE COMPULSORY.
(SECTION A) [7 X 2 = 14]

- Write short note on Compiler and Interpreter.
- Define the term Search engine & Web browser with examples.
- Define Cryptocurrency with its applications.
- Compare Quantum computing with Classical computing.
- Differentiate between Internet and WWW.
- Explain Emerging technology with example.
- Briefly explain any two application areas of block chain.

[5 X 7 = 35]

- How IOT is different from Industrial IOT?
- Describe cloud computing and their benefits.
- Differentiate between Green computing Vs Grid computing.
- Discuss crawling, indexing and ranking in context to search engines.
- On what basis can a city be called a smart city? Illustrate the role of technology for making a city to be a smart city.
- Build a flow chart to calculate factorial of a number.

[3X7= 21]

- Discuss the need of brain-computer interface. Explain its features and limitations.
- Explain the role of operating system in process management, file management, device management and memory management.
- Discuss AWS. Describe any five services of AWS.
- What do you mean by Memory? Classify memory hierarchy in terms of speed, size and cost.

(Following paper code and roll no. to be filled by student at the answer books)

PAPER CODE: BMC 102	ROLL NO.																			
---------------------	----------	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

BBS COLLEGE OF ENGINEERING & TECHNOLOGY

PRAYAGRAJ
SECOND SESSIONAL EXAM (2025-2026)

COURSE: MCA **SEM: 1st**
SUBJECT: Problem Solving Using C
MAX MARKS: 70 **MAX TIME: 2:00 HR**

NOTE: 1. QUESTION PAPER CONTAINS THREE PARTS.
2. ALL SECTIONS ARE COMPULSORY.

(SECTION A) **[7 X 2 = 14]**

Q1. ATTEMPT ALL FROM THE FOLLOWING:

- Briefly differentiate between while and do-while loop.
- Describe rules of declaring identifiers.
- There is a two-dimensional array of type integer. Write the statement to display the base address of the array.
- What is the use of return statement?
- What will be the output of the following code?

```
for (i=1; i<10;i++) ;  
printf ("%d", i);
```

f. Differentiate between structure & union.

g. Though we can write our program without File handling, what is the need of file handling in C?

(SECTION B)

[5 X 7 =35]

Q2. ATTEMPT ANY FIVE OF THE FOLLOWING:

- Discuss any five library functions used in C language.
- What do you mean by a storage class? Describe various storage classes with proper example.
- Write a C program to enter marks of 5 subjects of a student and find its division (first, second, third and fail).
- On successful execution of the below code predict the result, if the address of variable 'a' is 200.
void main()
{
float a, *b;
b = &a;
b = b+3;
printf ("%d", b);
}
- Write a program in C to count the number of characters in a string.
- Explain the use of rectangle(), circle() and line() function.

PAPER CODE	BVIC105	ROLL NO					
------------	---------	---------	--	--	--	--	--

SECOND SESSIONAL EXAM (2025-2026)

SEM: 1^{re}

MAX TIME: 02:00 HR

MAX TIME: 02:00 HR

2. ALL SECTIONS ARE COMPULSORY

 $[7 \times 2 = 14]$

- Write short note on RISC?
- Define micro-operation and micro code.
- What do you mean by page fault?
- What are the different types of instruction formats?
- What is non-restoring method?
- What is pipelining?
- Define miss ratio.

[5 X 7 = 35]

Q2. ATTEMPT ANY FIVE OF THE FOLLOWS:

- Define the term Memory Hierarchy in computer system.
- Compare microinstruction and microprogram.
- Discuss the need of virtual memory. Demonstrate its implementation using paging.

- (SECTION C)

[3X7= 21]

- What is an Associative memory? What are its advantages and disadvantages?
- Show the multiplication process using Booth's algorithm when the following numbers are multiplied: $(-21) \times (-11)$.
- Write a program to evaluate the arithmetic statement.
 - $P = ((X - Y + Z) * (A \wedge B)) / (C \wedge D + E)$
 - $X = A + B * [C * D - E * (F + G)]$

By using (i) Two address instructions
 (ii) One address instruction
 (iii) Zero address instructions

- d. Calculate the page fault for a given string with the help of LRU, FIFO & Optimal page replacement algorithm, Size of frame = 4 and string 1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6.